Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Full-stack software engineer with 2+ years building **TypeScript**, **React**, Python, real-time interfaces, APIs, and reliable distributed workflows for data-intensive and autonomous-system applications.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Operating Systems, Analysis of Algorithms, Database Systems, Software Engineering

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Structures, Computer Networks, Cloud Computing, Web Technologies

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *LA, California*
- Architected a multi-tenant platform on Azure SQL and Blob Storage with row-level security and normalized schemas for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds.
- Restructured fault-tolerant asynchronous orchestration using FastAPI workers, explicit job-state transitions, and failure isolation, processing 1K+ jobs while increasing throughput threefold under concurrent production workloads with reliable recovery.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *LA, California*
- Built a React and Flask product serving 150+ instructors and 10K+ daily requests, translating grading workflows into reliable interfaces and reducing feedback turnaround time by 60% at scale.
- Designed concurrent batch processing and secure file ingestion through AWS RDS and **REST APIs**, reliably processing 1K+ files per batch with fault-tolerant execution under peak workloads at scale.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles*
- Built an end-to-end Python pipeline processing 100K+ autonomous-driving trajectory logs, improving long-horizon planning by 18% through structured modeling, repeatable testing, and detailed simulation analysis at research scale.
- Architected a spatial-temporal encoder integrating HD-map geometry and vehicle histories through multi-head attention, improving closed-loop planning stability by 15% while validating behavior across simulation scenarios at scale.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*
- Engineered cross-platform microservices with automated builds and 20+ unit and integration suites using Jenkins, PyTest, and Mockito across Windows, Linux, and macOS release environments with repeatable nightly reporting.
- Troubleshot large-scale **API integrations** using OOP patterns, AWS Glue, S3, and SQL, improving response time by 30%, pipeline efficiency by 68%, and maintainability across production workflows at scale.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*
- Developed a React and Node.js resource-management platform with WebSockets, REST APIs, and eight microservices supporting authentication, billing, notifications, and real-time synchronization of 2M+ records across distributed locations.
- Automated Docker-based builds, testing, and staged deployments using Jenkins CI/CD, reducing deployment time by 50% while maintaining modular services and repeatable production-release workflows with automated quality gates.

Skills

Languages: TypeScript, JavaScript, Python, C++, Java, SQL, Bash
Frontend: React, Next.js, component-based UI, HTML, CSS, responsive design, accessibility, real-time dashboards
Backend & Data: Node.js, FastAPI, Flask, REST APIs, WebSockets, microservices, PostgreSQL, SQL Server, Azure SQL, MongoDB
Cloud & Infrastructure: AWS, Azure, GCP, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Git, Linux
Engineering: distributed systems, operating systems, data modeling, unit testing, integration testing, code review, monitoring, debugging, performance tuning, Agile/Scrum

Projects

RailOpsConsole | TypeScript, React, telemetry processing, operator interface
- Built and tested a TypeScript and React operator console that rejects out-of-order telemetry, detects stale and faulted vehicles, blocks unsafe commands, and summarizes real-time fleet state at scale.

CPU-scheduling Simulator | Live | C++, React, JavaScript
- Implemented six scheduling algorithms with React controls, dynamic execution metrics, and Gantt visualization across 1K+ process combinations, reducing average turnaround time by 25% in benchmark scenarios at scale.

FinNexus | Live | Next.js, TypeScript, Kubernetes, MongoDB
- Built a full-stack TypeScript platform supporting 50K+ users and 1M+ transactions with real-time synchronization, advanced filtering, secure APIs, and Kubernetes-backed deployment across reliable customer workflows at scale.